

RBSE · BSER AJMER · CLASS 12 · BOARD EXAM 2027

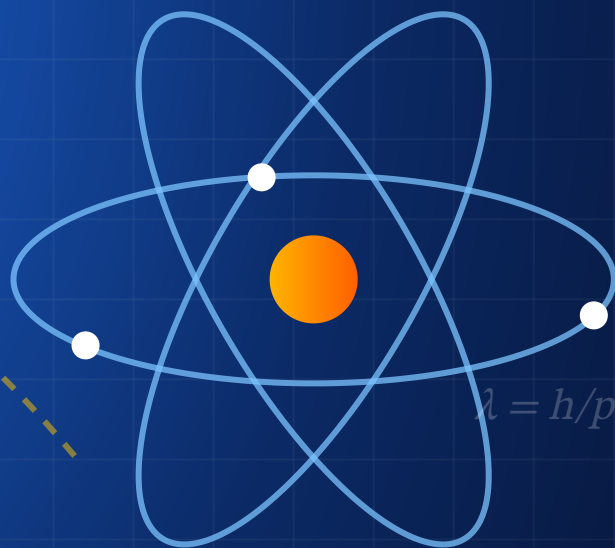
$$1/f = 1/v - 1/u$$

PHYSICS

BOARD EXAM COMPLETE GUIDE

Notes · Formulas · Diagrams · Most-repeated board questions · Answer-writing tips · 5 Model Papers in the new 56-mark pattern

ENGLISH MEDIUM



$$E = h\nu$$

All 14 chapters · notes + diagrams

$$V = IR$$

Chapter-wise weightage (56 marks)

PYQ analysis of official papers

Step-marked derivations

5 Model Papers + answers

30–45 day revision map

$$\varepsilon = -d$$



VS SIR

Vijay Kumar Sharma
20+ years · Physics expert · Kota

RBSE Class 12 Physics: Board Exam Complete Guide

Rajasthan Board (BSER, Ajmer) · Senior Secondary · Theory 56 marks

by VS Sir (Vijay Kumar Sharma)

20+ years teaching Physics, Kota

First edition: 2026 · Language: English

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About the questions: Questions marked [RBSE year] are previous-year questions from the official question papers published by the Board of Secondary Education, Rajasthan (rajeduboard.rajasthan.gov.in), given here in summarised form. All other questions, notes and the 5 model papers are original material written by the author in the board pattern. This book is an independent study aid and is not published or endorsed by BSER.

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ABOUT THE AUTHOR

VS Sir

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- 20+ years of teaching Physics in Kota – Class 12 boards, NEET and JEE
- Knows exactly how board examiners mark: keywords, diagrams and step marks
- Author of the NEET + JEE Physics 30-Day Rapid Revision series (Class 11 & 12, English & Hindi)

A message from VS Sir

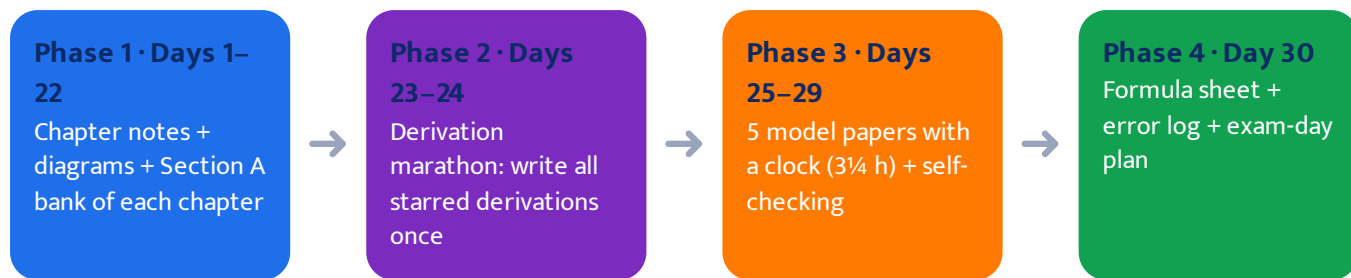
Dear student,

The RBSE Physics paper is very predictable if you know where to look. More than half of the marks come from Optics, AC, EMI, Magnetism and Electrostatics, and the same derivations return year after year. This book is built on the official BSER question papers: it shows you what is asked, how much it is worth, and exactly how to write the answer to get full marks.

Follow the revision map, write every derivation once with your own hand, and solve the five model papers with a clock. 56 out of 56 is possible.

— VS Sir

Your 30–45 day revision map



Have 45 days? Add a second round

- Days 31–35: second round of all derivations (Sections C, D)
- Days 36–40: re-solve every wrong question from the 5 model papers
- Days 41–44: official BSER papers of 2023, 2024, 2025 with a clock
- Day 45: formula sheet + diagrams only

How to study each chapter (2–3 hours)

1. **Read** the chapter snapshot: marks, what was asked, what is likely.
2. **Learn** definitions word-for-word – 1-mark answers need exact keywords.
3. **Write** each derivation once on paper with the diagram – reading is not enough.
4. **Practise** the question bank: MCQ and blanks in 1 min each; short answers in 5 min.
5. **Check** with the answers; mark every mistake in an error log and revisit on Day 30.

Golden rule of board Physics

Diagram + formula + final result in a box = full marks. Examiners look for the labelled diagram and the boxed answer first.

30-day plan (tick each day)

Daily target: 2–3 hours. Heavy chapters (Optics, AC, Magnetism) get extra days.

Day 1 Ch 1 Electric charges & fields – I	Day 2 Ch 1 Dipole, flux, Gauss law	Day 3 Ch 2 Potential & capacitance	Day 4 Ch 3 Current, Ohm, cells	Day 5 Ch 3 Kirchhoff, Wheatstone bridge
Day 6 Ch 4 Biot–Savart, Ampere	Day 7 Ch 4 Force, galvanometer	Day 8 Ch 5 Magnetism & matter	Day 9 Ch 6 EMI: Faraday, Lenz, inductance	Day 10 Ch 7 AC: rms, R/L/C circuits
Day 11 Ch 7 LCR, resonance, power, transformer	Day 12 Ch 8 EM waves	Day 13 Ch 9 Mirrors, refraction, TIR	Day 14 Ch 9 Lenses, lens maker, prism	Day 15 Ch 9 Microscope, telescope
Day 16 Ch 10 Huygens, YDSE	Day 17 Ch 10 Diffraction, polarisation	Day 18 Ch 11 Photoelectric effect, de Broglie	Day 19 Ch 12 Atoms: Bohr model	Day 20 Ch 13 Nuclei: BE, fission, fusion
Day 21 Ch 14 Semiconductors, p–n junction	Day 22 Ch 14 Rectifiers, special diodes	Day 23 Final Derivation marathon: Electricity & Magnetism	Day 24 Final Derivation marathon: AC, Optics, Modern	Day 25 Final Model Paper 1 (3¼ h)
Day 26 Final Model Paper 2	Day 27 Final Model Paper 3	Day 28 Final Model Paper 4	Day 29 Final Model Paper 5	Day 30 Final Formula sheet + exam-day plan

Paper pattern (as in the 2025 board paper)

Subject code SS-40 · Time **3 hours 15 minutes** · Maximum marks **56** · **18** numbered questions · all compulsory
· internal choice only in Q14–18 · if Hindi and English differ, the Hindi version is valid.

Section	Q. no.	Type	Items	Marks each	Total
अ / A	1	Multiple choice (MCQ)	18	$\frac{1}{2}$	9
अ / A	2	Fill in the blanks	10	$\frac{1}{2}$	5
अ / A	3	Very short answer	10	1	10
ब / B	4–13	Short answer	10	$1\frac{1}{2}$	15
स / C	14–16	Long answer (with OR)	3	3	9
द / D	17–18	Essay type (with OR)	2	4	8
Total			53		56

How the 100 marks are made

Theory paper

56

Practical

30

Sessional
(school)

14

Time plan (195 min)

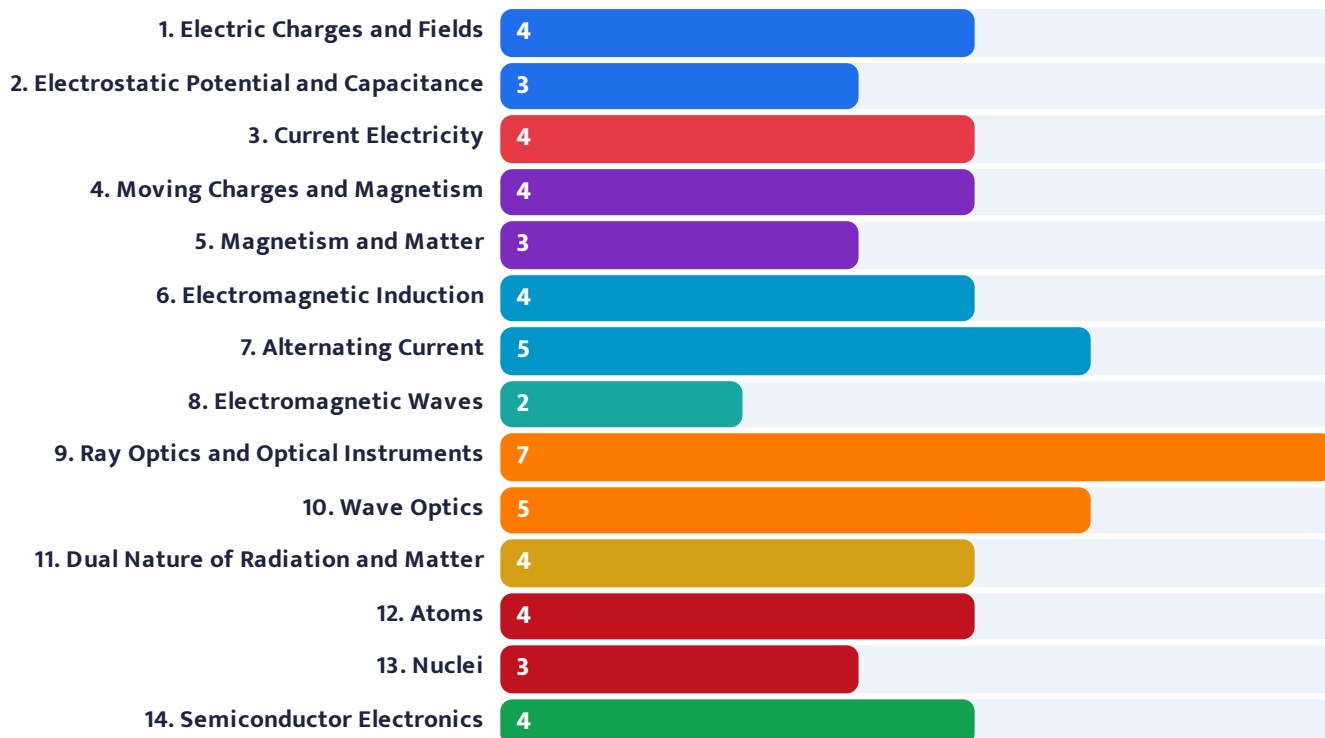
- Section A (28 items): 50 min
- Section B ($10 \times 1\frac{1}{2}$): 60 min
- Section C (3×3): 35 min
- Section D (2×4): 40 min
- Revision: 10 min

Pattern has changed recently

2023 and 2024 papers had 20 questions; 2025 had 18 questions with $\frac{1}{2}$ -mark MCQs and choice in Q14–18. The 2026 CWSN paper also used 18 questions. Always check the latest BSER model paper for your exam year.

Chapter-wise weightage (56 marks)

Marks per chapter as per the RBSE syllabus blueprint. Spend your time in this ratio.



Unit view: top 5 units = 42 of 56 marks (75%)



Note

Chapter marks are from the RBSE blueprint for 2025-26 (sum = 56). Websites quote slightly different splits for Current Electricity and Semiconductors ($\pm 1-2$ marks); the board can change the blueprint each year, so confirm with the latest syllabus on rajeduboard.rajasthan.gov.in.

What the board keeps asking

Topic-wise analysis of the official BSER Physics papers (SS-40). ✓ = asked in that year (any section). *2026 = CWSN paper of the same pattern.

Repeated topic	Ch	2023	2024	2025	2026*
Electric field of dipole (axis / equatorial)	1	✓	✓	✓	✓
Gauss's law: infinite plane / line charge	1	–	✓	✓	✓
Potential of point charge, PE of charge system	2	✓	–	✓	–
Wheatstone bridge (Kirchhoff)	3	✓	✓	✓	–
EMF, internal resistance, cells in series	3	–	✓	✓	–
Biot–Savart: field on axis of loop	4	–	✓	✓	✓
Ampere's law: wire / solenoid	4	✓	–	✓	✓
Faraday, Lenz, self / mutual inductance	6	✓	✓	✓	✓
AC: rms, reactance, LCR, resonance, power	7	✓	✓	✓	✓
Transformer / AC generator	7	✓	✓	–	✓
EM waves: properties, spectrum	8	✓	✓	✓	–
Mirror formula, prism, lens maker, microscope	9	✓	✓	✓	✓
Huygens' principle (Snell's law), YDSE	10	✓	✓	✓	✓
Photoelectric effect, Einstein equation	11	✓	✓	✓	✓
Bohr model, energy of H-atom	12	✓	✓	✓	✓
BE/nucleon graph, fission, fusion, mass defect	13	✓	✓	✓	–
Energy bands, p-n junction, rectifier	14	✓	✓	✓	✓

🎯 Conclusion

Every topic in this table came at least twice in the last three regular papers. Learn these derivations and numericals first – they cover almost all of Sections B, C and D.

Contents

1	Electric Charges and Fields Charge & its properties · Coulomb's law · Electric field & field lines · Electric dipole	4 marks	10
2	Electrostatic Potential and Capacitance Potential & potential energy · Equipotential surfaces · Potential of a dipole · Conductors & dielectrics	3 marks	22
3	Current Electricity Drift velocity & mobility · Ohm's law & resistivity · emf & internal resistance · Cells in series/parallel	4 marks	34
4	Moving Charges and Magnetism Lorentz force · Circular motion in B · Biot-Savart law · Ampère's law & solenoid	4 marks	46
5	Magnetism and Matter Bar magnet & field lines · Torque & PE of dipole · Gauss's law · M, H, χ , μ_r	3 marks	60
6	Electromagnetic Induction Magnetic flux · Faraday's & Lenz's laws · Motional emf · Eddy currents	4 marks	71
7	Alternating Current AC generator · rms value · R, L, C circuits · Series LCR & resonance	5 marks	82
8	Electromagnetic Waves Displacement current · Maxwell's idea · Nature of EM waves · $c = 1/\sqrt{(\mu_0\epsilon_0)}$	2 marks	94
9	Ray Optics and Optical Instruments Mirror formula · Refraction & TIR · Lens maker's formula · Power & combination	7 marks	103
10	Wave Optics Wavefront & Huygens · Coherent sources · Young's double slit · Single-slit diffraction	5 marks	118
11	Dual Nature of Radiation and Matter Work function · Photoelectric effect · Stopping potential · Einstein's equation	4 marks	128
12	Atoms α -scattering & nuclear model · Closest approach · Hydrogen spectrum · Bohr's postulates	4 marks	138
13	Nuclei Size & density · Isotopes, isobars · Mass defect · B.E. per nucleon	3 marks	148
14	Semiconductor Electronics Energy bands · n- and p-type · p-n junction · Diode & rectifiers	4 marks	158
①	Model Paper 1 Full paper · 18 questions · 56 marks · 3 h 15 min · with marking scheme	56 marks	169
②	Model Paper 2 Full paper · 18 questions · 56 marks · 3 h 15 min · with marking scheme	56 marks	184
③	Model Paper 3 Full paper · 18 questions · 56 marks · 3 h 15 min · with marking scheme	56 marks	198
	Model Paper 4 Full paper in the 2025 pattern · SS-40 · 3¼ h · marking scheme + analysis	56 marks	214
	Model Paper 5 Full paper in the 2025 pattern · SS-40 · 3¼ h · marking scheme + analysis	56 marks	228
★	Final Toolkit: Formulas, Derivations, Errors and Exam-Day Plan 14-chapter formula sheet with units · derivation checklist · errors log · exam-day plan	Day 30	241

CHAPTER

1

Electric Charges and Fields

17

Day 1-2

Board marks: 4

PYQ frequency ★★★

RBSE Class 12

Charge & its properties

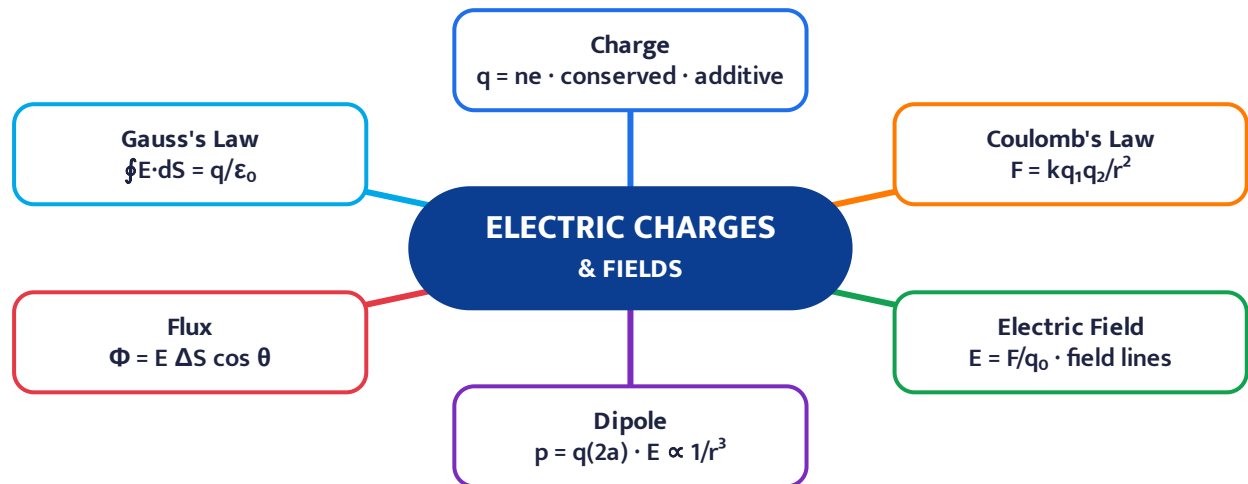
Coulomb's law

Electric field & field lines

Electric dipole

Electric flux

Gauss's law & applications



Chapter at a glance



Board snapshot

- **Marks: 4** (of 56). Usually one MCQ / fill-in (½ each) + one 1½-mark numerical **or** a 3-mark derivation (Section C, with OR).
- **2025:** MCQ – SI unit of electric flux; MCQ – force when $r \rightarrow 2r$; fill-in – dipole field $\propto 1/r^3$; Section B – λ of an infinite line charge ($E = 4.5 \times 10^4$ N/C at 4 cm).
- **2024:** MCQ – flux through a Gaussian sphere when its radius is tripled; fill-in – field lines of $+q$ are radially outward; Section C – dipole field on the equatorial line **OR** field of an infinite plane sheet (Gauss).
- **2023:** MCQ – permittivity of free space ϵ_0 ; fill-in – field inside a charged spherical shell; VSA – field lines, dipole field. **2026 (CWSN):** Section C – dipole field **OR** infinite plane sheet. **2020:** electron orbiting an infinite line charge.
- **Most likely in 2027:** (1) dipole field – axial / equatorial (3 marks); (2) Gauss's law with infinite line charge or plane sheet; (3) ½-mark items on flux unit, Coulomb's law ratio, field inside a shell.

1.1 Electric Charge and its Basic Properties

Electric charge is an intrinsic property of matter due to which it produces and experiences electric and magnetic effects. Charges are of two kinds – **positive** and **negative**; **like charges repel and unlike charges attract**. When two bodies are rubbed, **electrons are transferred** from one to the other: the body that loses electrons becomes positive. SI unit of charge: **coulomb (C)**.

**Three basic properties**

- **Additivity:** total charge = algebraic sum, $q = q_1 + q_2 + \dots$ (charge is a **scalar**).
- **Conservation:** the total charge of an isolated system remains constant; charge can be neither created nor destroyed.
- **Quantisation:** $q = \pm ne$, $n = 1, 2, 3, \dots$;
 $e = 1.6 \times 10^{-19} \text{ C}$ is the smallest free charge.

**Solved Example**

How many electrons must be removed from a body to give it a charge of $+3.2 \times 10^{-7} \text{ C}$?

Solution: $n = \frac{q}{e} = \frac{3.2 \times 10^{-7}}{1.6 \times 10^{-19}} = 2 \times 10^{12} \text{ electrons.}$

**Note**

For macroscopic charges (μC) quantisation is not noticeable – charge is treated as **continuous**. $1 \text{ C} \approx 6.25 \times 10^{18} \text{ electrons.}$

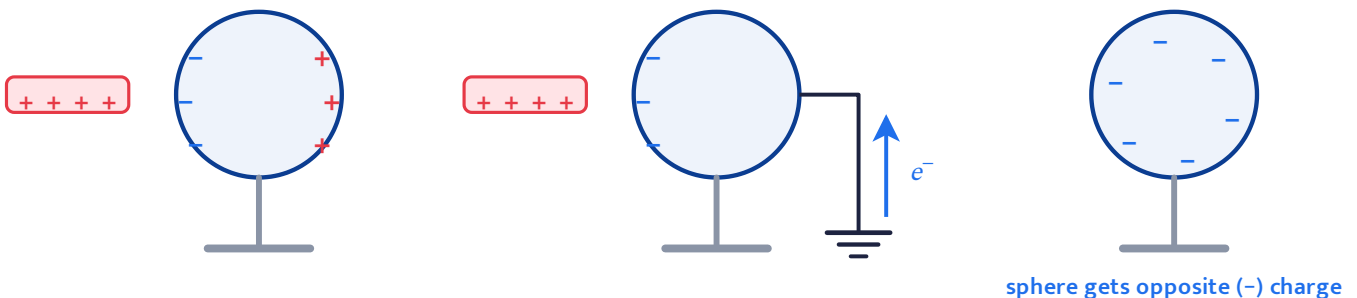
1.2 Conductors, Insulators and Charging by Induction

Conductors (metals, human body, earth) allow electric charge to move freely through them because they contain free electrons. **Insulators** (glass, plastic, nylon, porcelain) do not. Charge given to a conductor spreads over its **outer surface**. **Earthing (grounding)** connects a body to the earth, which acts as an infinite sink of charge. In **charging by induction**, a charged body redistributes the charges of a nearby conductor **without touching it**; the conductor finally gets a charge **opposite** to that of the inducing body.

1. Bring a +ve rod near

2. Earth the far side

3. Remove earth, then rod



Charging a metal sphere by induction – the charged rod never touches the sphere

**Common Trap**

- A charged body also attracts a **neutral** body (due to induction). So **attraction is not a sure test** of charge – **repulsion is**.
- In induction, remove the earth connection **first** and the rod **afterwards**; otherwise the sphere becomes neutral again.

1.3 Coulomb's Law and Superposition

Coulomb's law: the force of attraction or repulsion between two stationary point charges is **directly proportional to the product of their magnitudes** and **inversely proportional to the square of the distance** between them; it acts along the line joining the two charges.

Coulomb's law

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

Permittivity of free space (vacuum)

$$\vec{F}_{21} = k \frac{q_1 q_2}{r^2} \hat{r}_{21} = -\vec{F}_{12}$$

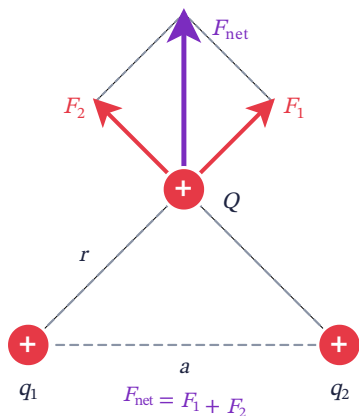
Vector form: force on q_2 due to q_1 ; $\hat{r}_{21}$ points from 1 to 2

$$F_m = \frac{F_0}{K}, \quad K = \frac{\epsilon}{\epsilon_0}$$

In a medium: $K =$ **dielectric constant** (relative permittivity)

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n}$$

Superposition: net force = vector sum of forces due to each charge separately



Superposition: forces add as vectors

Solved Example

Charges $+2 \mu\text{C}$ and $-3 \mu\text{C}$ are 30 cm apart in air. Find the force between them.

Solution: $F = \frac{kq_1 q_2}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 3 \times 10^{-6}}{(0.3)^2} = 0.6 \text{ N}$, attractive.

VS Sir's Exam Tip

Ratio questions: $F \propto \frac{q_1 q_2}{r^2}$. Distance doubled $\Rightarrow F/4$; distance halved $\Rightarrow 4F$; medium $K \Rightarrow F/K$.

1.4 Electric Field

The **electric field** at a point is the force experienced by a **unit positive test charge** placed at that point:

$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$. The test charge is taken very small so that it does not disturb the source charges. It is a **vector**; SI

unit N C^{-1} ($= \text{V m}^{-1}$). A charge q in a field experiences $\vec{F} = q\vec{E}$ – along $\vec{E}$ for $+q$ and **opposite** to $\vec{E}$ for $-q$.

Electric field

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Point charge: radially outward for $+q$, inward for $-q$; $E \propto 1/r^2$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots = \sum_i \frac{kq_i}{r_i^2} \hat{r}_i$$

System of charges – superposition (vector sum)

1.5 Electric Field Lines

An **electric field line** is a curve drawn in such a way that the **tangent at any point gives the direction of the electric field** at that point. The relative closeness of the lines shows the strength of the field.